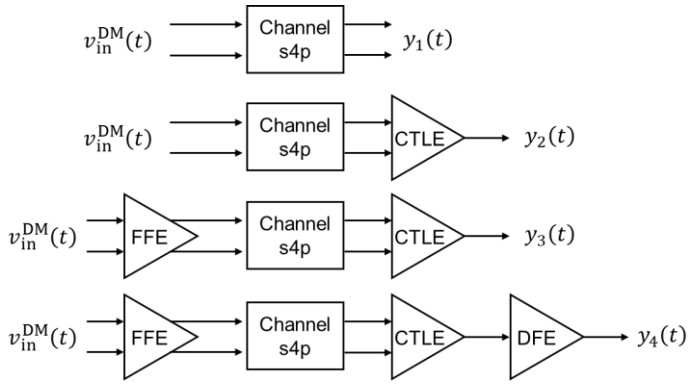
Homework: Differential channel simulation and equalizer design

An S4P is provided to you which is the S parameter of a differential channel. Your task is to design CTLE, FFE, and DFE for this channel such that the post-DFE eye has EH > 120 mV and EW > 15 ps. The equivalent circuit at each source port is a resistor Z_0 = 50 Ω in series with an ideal voltage source having bit rate 40 Gbps, risetime 5 ps, and *V*pp = 2 V. Each load port is also terminated by 50 Ω. Design the CTLE first, then FFE, and then finally DFE, as shown below.



In your report, provide the following:

1. A figure comparing the channel's S_{21} versus S_{dd21}
2. Channel-only eye
3. Channel+CTLE eye
4. FFE+channel+CTLE eye
5. FFE+channel+CTLE+DFE eye
6. Comparison of output waveforms ($y_i(t)$) of cases 1~4
7. All the relevant parameters of your equalizers

Design constraints

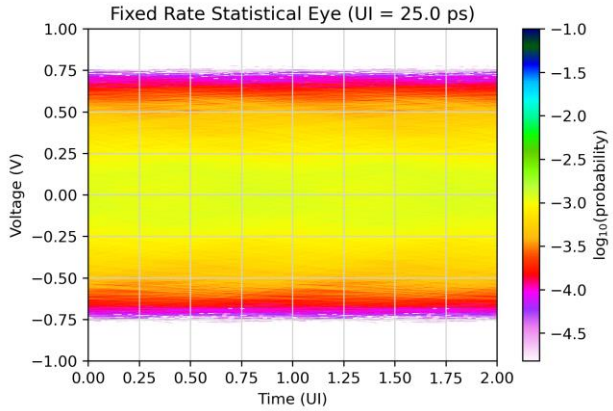
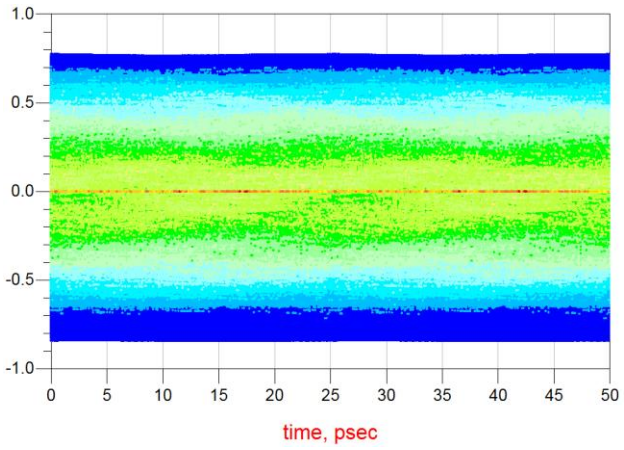
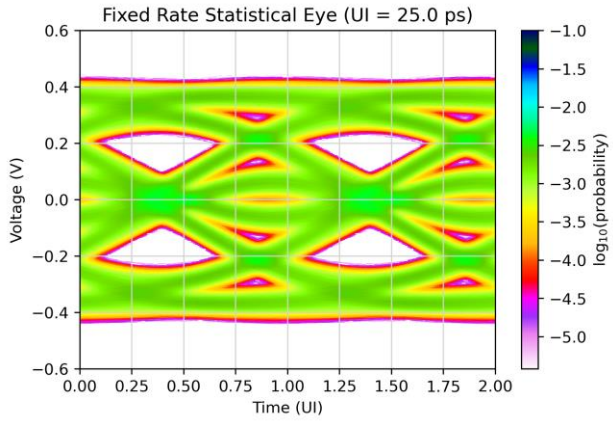
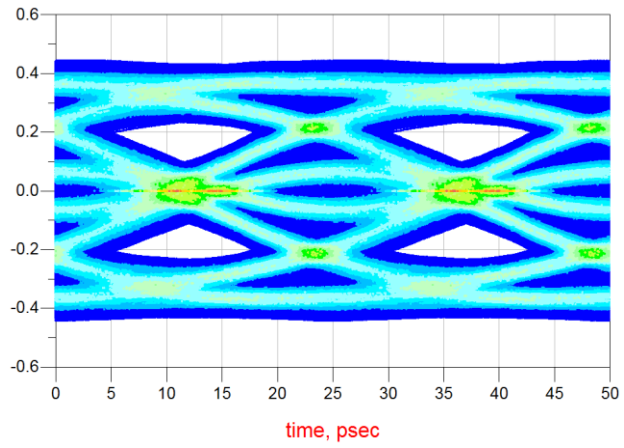
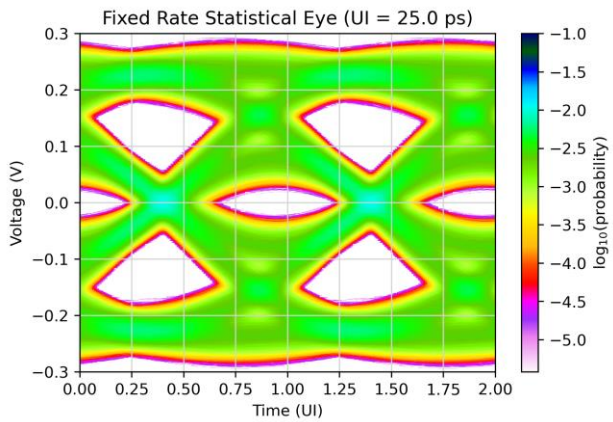
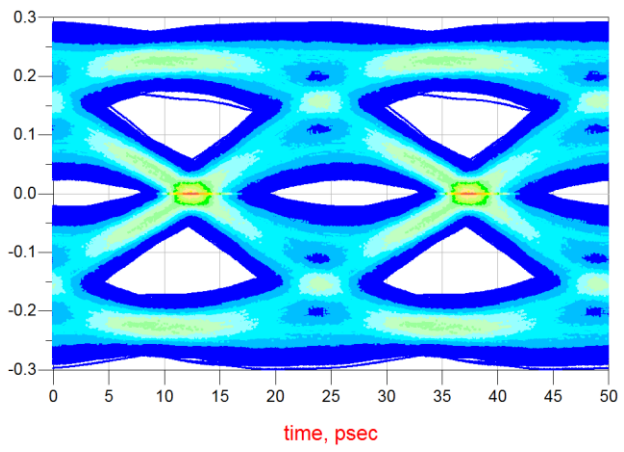
CTLE	FFE	DFE
Maximum three poles Highest gain ≤ 6 dB Highest pole ≤ 40 GHz	Maximum 5 taps $\sum_i \lvert c_i \rvert = 1$	Maximum 5 taps

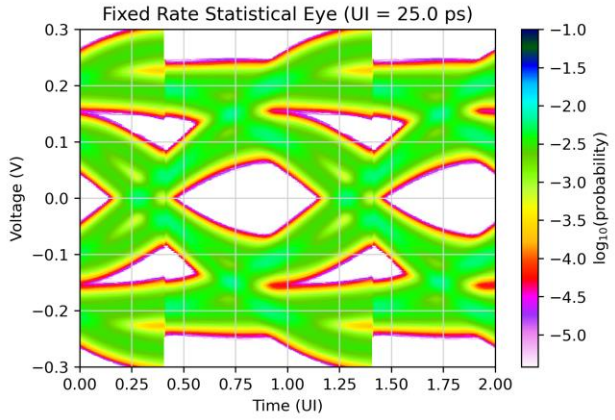
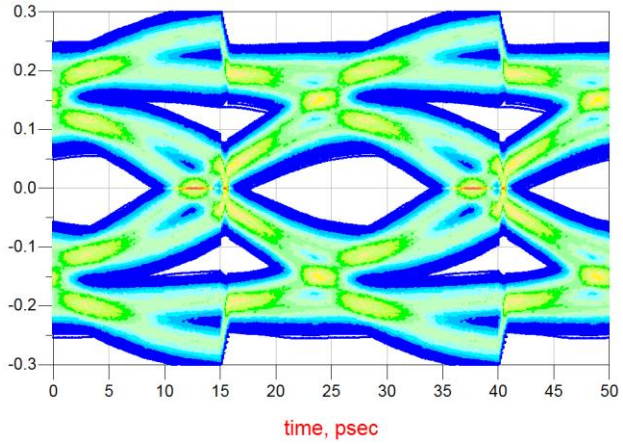
Note: the ZF solution of FFE is not always the optimal one, because you will have a DFE at Rx which can also cancel post-cursor ISI. You may look at the SBR and choose different FFE taps yourself.

Optional problem: Simulate the circuit in ADS ChannelSim to verify your design.

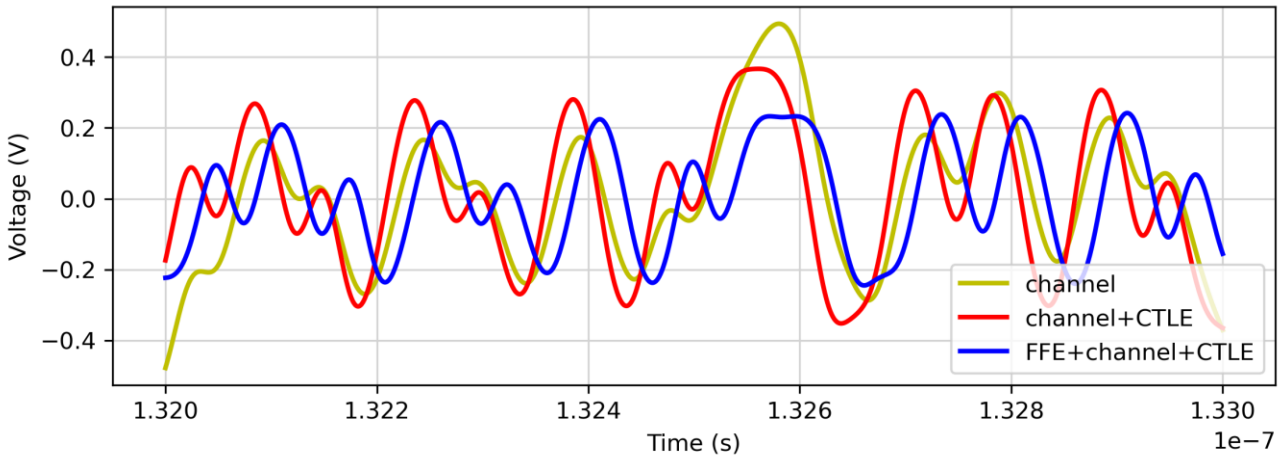
Appendix 1: Example answer

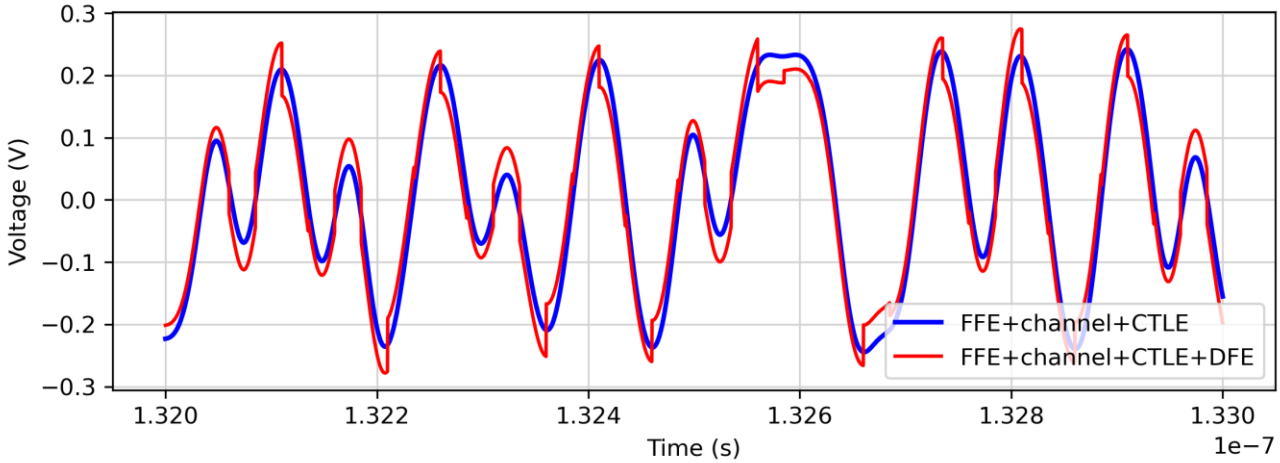
Nbp = 2^{18}

Python's eyes	ADS's eyes
Channel-only	
	
Channel+CTLE	
	
FFE+Channel+CTLE	
	

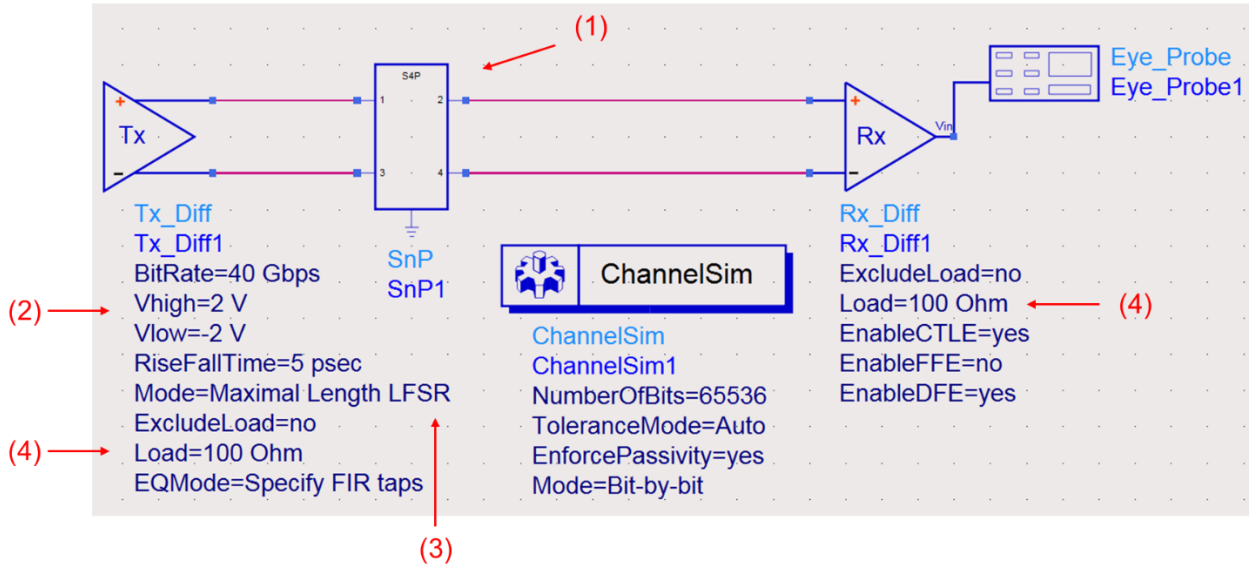
FFE+Channel+CTLE+DFE	
	

Waveforms





Appendix 2: ADS setting



(1) Be careful of port ordering: 13 at left, 24 at right

(2) In ADS, this Vhigh/Vlow setting refers to the DM input signal, i.e., this driver will never generate CM signal into the circuit. In this homework, the DM input swings between ±Vpp before impedance division.

(3) Remember to set # taps (register length) to 16 or larger.

(4) Terminate by $Z_d = 2Z_0$, such that the DM is matched. The CM is also matched because ADS will automatically terminate CM by Load/4. (See its document)

In the Rx's DFE setting, remember to set Slicer output to 1/-1.

